

CHAPTER 2 · CONCEPT 1

Exponential Notation

A power is a short way to write repeated multiplication. Learn the words base and exponent, how to turn products into powers, and how to write a number as a product of prime powers.

Base, exponent and value

The idea: n^a means **n multiplied by itself a times**. We call **n** the **base** and **a** the **exponent** (or power).

$$5^4 = 5 \times 5 \times 5 \times 5 = 625 \quad | \quad \text{base} = 5, \text{exponent} = 4$$

Read 5^4 as "5 raised to the power 4" or "5 to the power 4". Special names: a^2 is "squared", a^3 is "cubed".

Writing a product as a power

If the same factor repeats, count how many times and write it as a power. Different factors get their own powers.

$$2 \times 2 \times 2 \times a \times a = 2^3 \times a^2 = 2^3a^2$$

Powers with negative bases — watch the sign

- A negative base raised to an **even** power is **positive**: $(-4)^2 = 16$.
- A negative base raised to an **odd** power is **negative**: $(-4)^3 = -64$.

Prime factorisation in exponential form

Break the number into primes, then write each repeated prime as a power.

$$648 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 2^3 \times 3^4$$

Prerequisite refresher: A "factor" is a number that divides exactly. "Prime factorisation" writes a number using only prime factors (2, 3, 5, 7, ...).

Where marks slip away

Trap 1 — Power $\neq$ multiply by the exponent. $5^4 = 625$, not $5 \times 4 = 20$.

Trap 2 — Sign of a negative base. $(-2)^4 = +16$ (even power), but $(-2)^5 = -32$ (odd power). Count the power's parity.

Trap 3 — -2^4 vs $(-2)^4$. Without brackets, $-2^4 = -(2^4) = -16$. With brackets, $(-2)^4 = +16$. Brackets matter.

Trap 4 — Counting the primes. In prime factorisation, count each prime carefully — one extra or missing factor changes the exponent.

Slip 5 — Notation. Write the exponent small and raised: 2^3 . "23" means twenty-three.

Model answers — write them like this

Example A. Express $2 \times 2 \times a \times a \times a$ in exponential form.

Step 1: Count each repeated factor: 2 appears twice, a appears three times.

Step 2: Write each as a power: 2^2 and a^3 .

$$\therefore 2 \times 2 \times a \times a \times a = 2^2 a^3$$

Example B. Express 648 as a product of prime powers.

Step 1: Divide by primes: $648 = 2 \times 324 = 2 \times 2 \times 162 = 2 \times 2 \times 2 \times 81$.

Step 2: $81 = 3 \times 3 \times 3 \times 3$, so $648 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$.

Step 3: Write repeated primes as powers.

$$\therefore 648 = 2^3 \times 3^4$$

Example C. Find the value of $(-3)^2 \times (-5)^2$.

Step 1: $(-3)^2 = 9$ (even power $\rightarrow$ positive).

Step 2: $(-5)^2 = 25$ (even power $\rightarrow$ positive).

Step 3: $9 \times 25 = 225$.

$$\therefore (-3)^2 \times (-5)^2 = 225$$

Example D. Find the value of $(-2)^5 \times (-10)^6$.

Step 1: $(-2)^5 = -32$ (odd power $\rightarrow$ negative).

Step 2: $(-10)^6 = 1000000$ (even power $\rightarrow$ positive).

Step 3: $-32 \times 1000000 = -32000000$.

$$\therefore (-2)^5 \times (-10)^6 = -32000000$$

Next: [try the worksheet](#), then [check the answer key](#).